

Introducing Arduino Voice Controlled Relay for Home Automation

*A Major Project Report
Submitted in partial fulfillment of the requirement for the degree of*

***Bachelor of Technology
In
Electronics and Communication Engineering
By***

Abhishek Brajvasi Scholar No. 20U01047
Arpeet Bhaisare Scholar No. 20U01041
Samarth Varshney Scholar No. 20U01024

Supervisors:
Dr. Bhupendra Singh Kirar



Department of Electronics and Communication Engineering
Indian Institute of Information Technology
Bhopal-462003 (India)
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Certificate

Department of Electronics and Communication Engineering Indian Institute of Information Technology Bhopal

It is certified that the work contained in the project report entitled “Introducing Arduino Voice Controlled Relay for Home Automation” by the following Students has been carried out under my/our supervision and that this work has not been submitted elsewhere for a degree.

Abhishek Brajvasi	20U01047
Arpeet Bhaisare	20U01041
Samarth Varshney	20U01024

Date:

Supervisor (Name & Signature)

This project report entitled “Introducing Arduino Voice Controlled Relay for Home Automation” submitted by the group is approved for the degree of Bachelor of Technology.
The viva-voce examination has been held on _____

Supervisor(s)

Examiner(s)

Declaration

IIIT Bhopal
Date

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We declare that we have properly and accurately acknowledged all sources used in the production of this report. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and can also evoke penalty action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Abhishek Brajvasi 20U01047 _____
Arpeet Bhaisare 20U01041 _____
Samarth Varshney 20U01024 _____

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Abhishek Brajvasi 20U01047
Arpeet Bhaisare 20U01041
Samarth Varshney 20U01024

Abstract

A voice-controlled home automation system uses an Android app and an Arduino mega board to control appliances with voice commands. The system uses three main components:

- Arduino Uno microcontroller
- Bluetooth module HC-05
- Mechanical relay

The system works by:

1. The user sends a voice command through the app.
 2. The Bluetooth module receives the command and passes it to the Arduino.
 3. The Arduino compares the command with predefined commands.
 4. The Arduino sends a command to the relays to govern the house equipment.
- The system is designed to be easy for non-technical people to carry, install, configure, run, and maintain.



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Introduction

In this project, we will control the home appliances by sending Voice Command from our Smartphone. We will use a Bluetooth module in our project that helps to connect and communicate with a Smartphone wirelessly. Also, we will use a Bluetooth app on our Smartphone, which is helping to send our voice Command from our smart phone to the project circuit.

➤ Project Concept

- Here we will control four different home appliances by voice Command.
- The key components of this project are Arduino, Bluetooth module, Relay module, LCD display, a Smartphone, and Android App.
- At first, we need to install the app on our Smartphone, which is easily available in the play store.
- This app receives our Voice command and sends it to the Bluetooth module wirelessly.
- The Arduino decodes this command from the Bluetooth module.
- Then Arduino sends a command to the Relays to control the home appliances.
- These four home appliances are switched on/off by eight different voice commands.
- The LCD module will display the status of the home appliances on or off.

Literature Review or survey

- Bluetooth based home automation system using cell phones:
- In Bluetooth based home automation system the home appliances are connected to the Arduino BT board at input output ports using relay.
- The program of Arduino BT board is based on high level interactive C language of microcontrollers; the connection is made via Bluetooth.
- The password protection is provided so only authorized user is allowed to access the appliances.
- The Bluetooth connection is established between Arduino BT board and phone for wireless communication.
- In this system the python script is used and it can install on any of the Symbian OS environment, it is portable.
- One circuit is designed and implemented for receiving the feedback from the phone, which indicate the status of the device.

Methodology & Work Description

- To introduce a voice-controlled relay for home automation using Arduino, you can follow these steps:

1. Install an Android app on your Smartphone.
2. Assemble the hardware.
3. Program the Arduino.
4. Connect the Arduino to a 9v battery or computer.
5. Define specific voice commands and map them to specific digital pins on the Arduino.
6. Connect the Arduino pin 5 to IN1 of the relay module and pin 6 to IN2.
7. The Arduino decodes the command from the Bluetooth module and sends a command to the relays.

- **The key components of this project are:**

- Arduino, Bluetooth module, Relay module, LCD display, Smartphone, Android app.
- The purpose of a home automation system is to provide ease to people to control different home appliances.

- We need to choose 8 different types of Voice Commands to control 4 devices.

Particular two voice commands are used to control a device. ,

- The “turn on light” command is used to Turn ON the Light, and the “turn off light” command is used to Turn OFF the Light.

- These Voice commands are used in **Arduino code**.

- The list of voice commands is below, which is used in Arduino code:

Voice commands	Device status	Control Relay (Relay Pin)
turn on light	Light On	Relay-1 (IN1)
turn off light	Light Off	
turn on fan	Fan On	Relay-2 (IN2)
turn off fan	Fan Off	
turn on TV	TV On	Relay-3 (IN3)
turn off TV	TV Off	
turn on pump	Pump On	Relay-4 (IN4)
turn off pump	Pump Off	
turn on all	All Device On	Relay-1, Relay-2 Relay-3, Relay-4 (IN1, IN2, IN3, IN4)
turn off all	All Device Off	

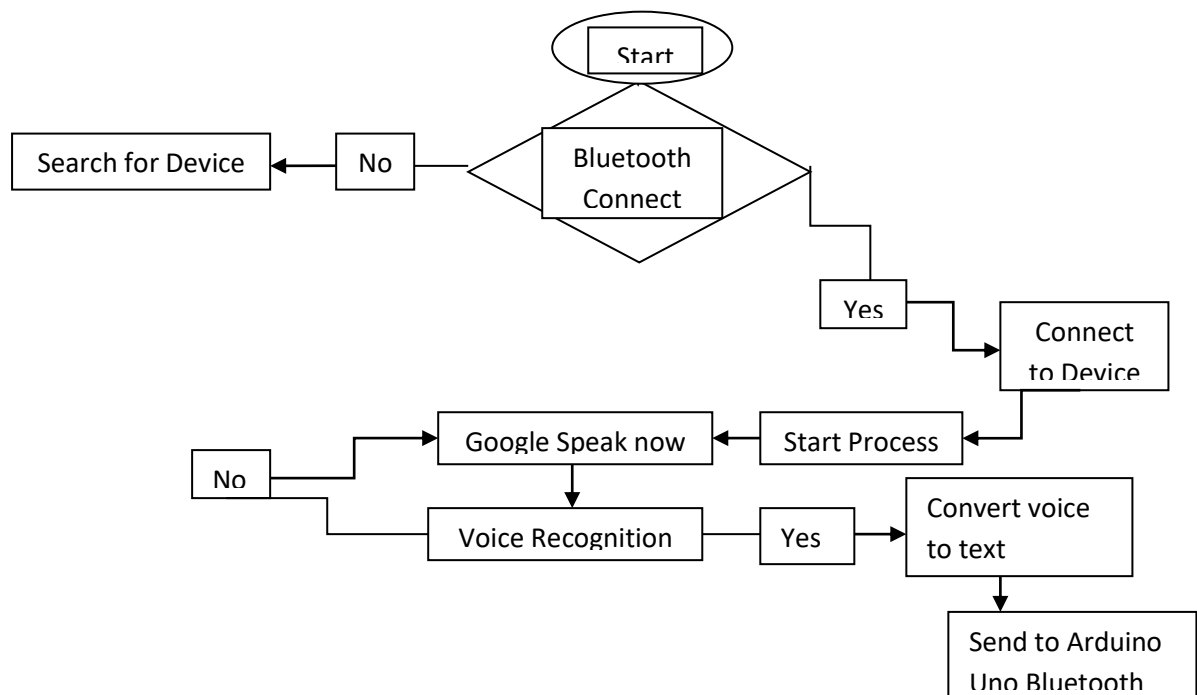
1. When we send a voice command through the App, then the Bluetooth module receives that command and passes it to the Arduino.
2. Now Arduino compares this command with the predefined Commands (which are defined in Arduino code).
3. If this command matches then Arduino sends a command to operate the relay module. We can see the Device status (on or off) on the 16×2 LCD Display Module.
4. When we send the “**turn on light**” voice command through the application.
5. Then the Arduino gets this command through the Bluetooth module.
6. Then the Arduino sends **Low (0)** input voltage to the **Input-1 (IN1)** pin of the relay module.
7. Now the relay is in **on mode**. So the device (light) will also **turn on**, which is connected to the relay-1 of the relay module.
8. At the same time, the “**D1 (Device 1) is ON**” status print on the 16×2 LCD Display Module.
9. When we send the “**turn off light**” voice command through the app.
10. Again the Arduino gets this command through the Bluetooth module.
11. This time the Arduino sends a **High (5v)** input voltage to the **Input-1(IN1)** pin of the relay module.
12. Now the relay is in **off mode**.
13. So the device (light) will also **turn off**, which is connected to the relay-1 of the relay module.
14. At the same time, the “**D1 (Device 1) is OFF**” status print on the 16×2 LCD Display Module.

15. In this way, we can control all relays of the 4 channel Relay module through the voice command

3

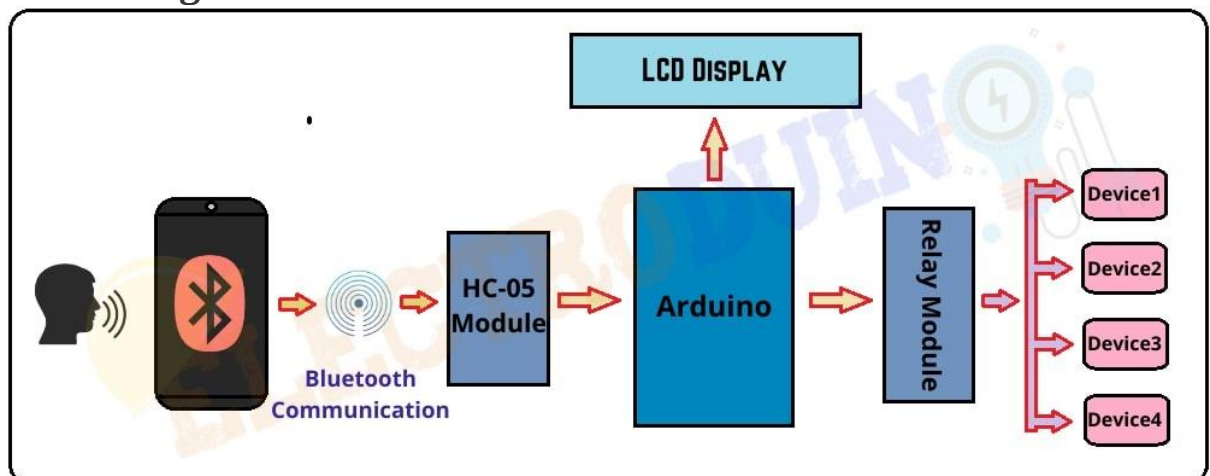
Proposed algorithm

It displays the respective stepwise algorithm of the application, which involves the complete processing of input data from the user, and this data involves the logical sequence that must be followed to make the program work. Following algorithm diagram of our project shows all the steps that are involved in the complete processing of the application and it also shows the Google voice recognition in the way it is connected to the application, and ultimately to the end spots i.e. Arduino Uno.

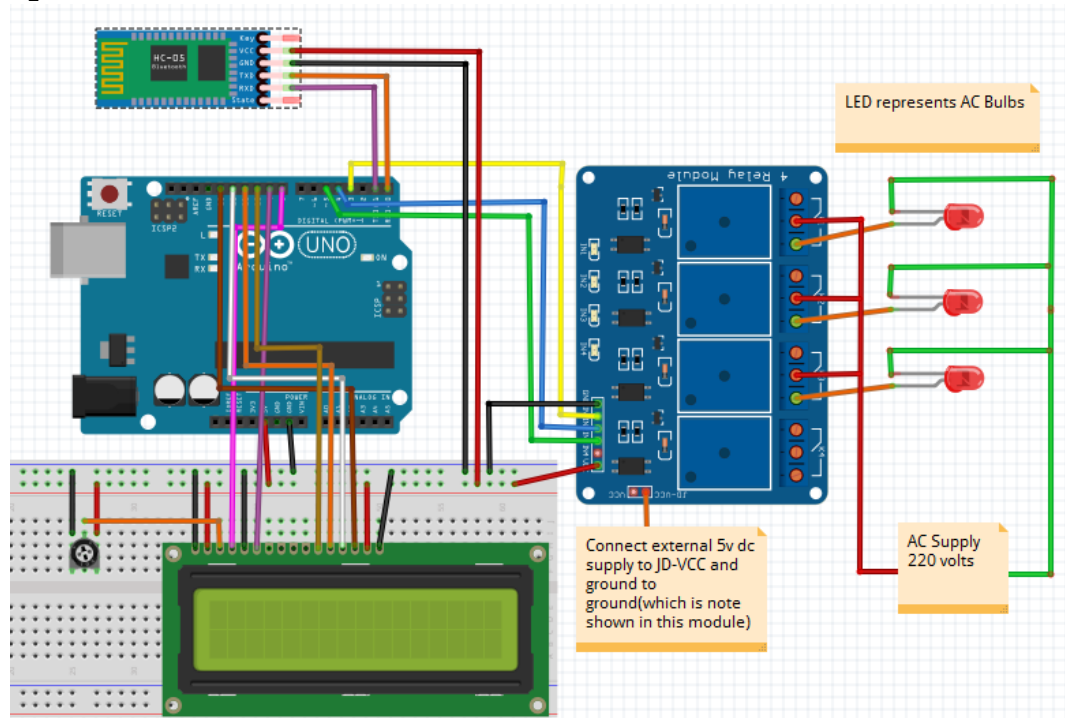


Proposed flowchart/ DFD/ Block Diagram

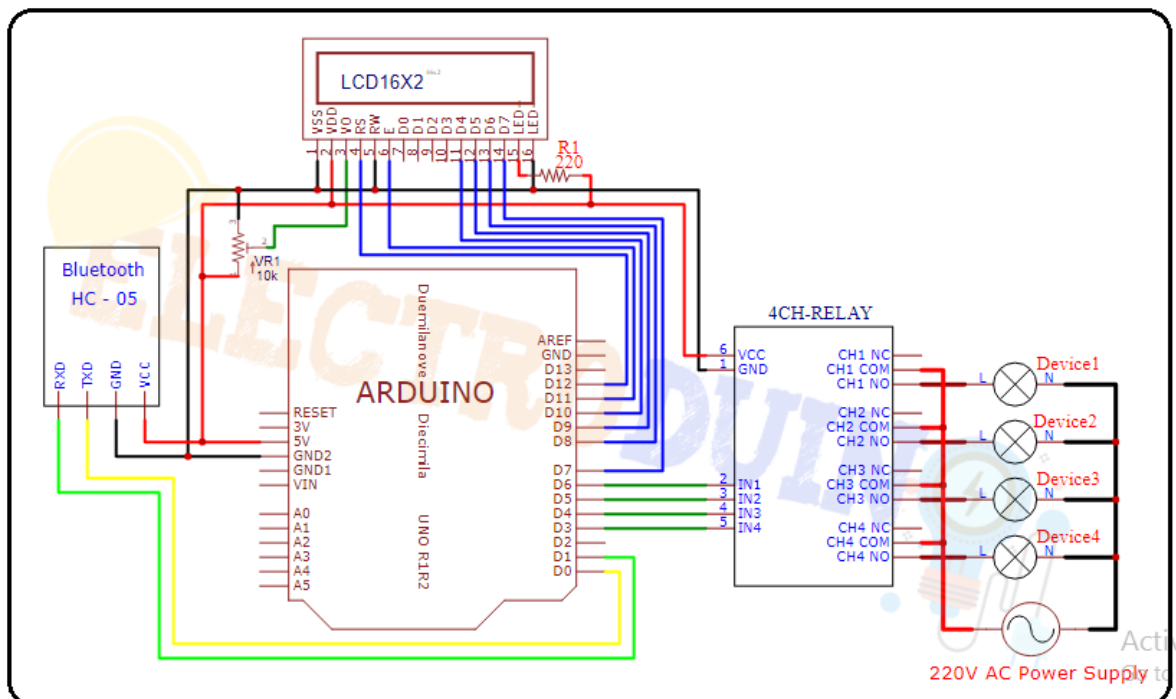
1] Block Diagram of Voice Controlled Home Automation



2] DFD



3] Circuit Diagram of Voice Control Home Automation using Arduino and Bluetooth Module HC-05





Tools & Technology Used

- To build a voice-controlled home automation system using Arduino, you need the following components:

- Arduino UNO
- HC-05 Bluetooth module
- 2-channel relay module (5v)
- 2 holders
- 2 bulbs
- 220v electrical wire with a 2-pin male socket
- Sun board piece
- Jumper wires
- Sketch fab software
- Hi-link 5V 3W power supply module
- Android app

The Arduino UNO controls appliances within the Bluetooth range of the house. The relay module is a trigger device that activates another appliance, such as a light fixture or garage door. The components are arranged on a breadboard using a circuit diagram stimulated on protease. The software part of the system is **programming on Arduino NANO and designing the Android mobile telephone operation for the system.**

1] Components Required

Components Name	Quantity
Arduino	1
HC-05 Bluetooth Module	1
4 Channel Relay Module	1
LCD display module	1
AC Bulb with Holder and Wire	4
PCB Prototyping board	1
12V DC Power supply	1
220V AC Power supply	1
Connecting wires	As required in the circuit diagram
Android Phone	1
“Bluetooth voice control for Arduino” App	Available in the play store

Implementation & Coding

- To implement a voice-controlled relay for home automation using Arduino, you can follow these steps:

1. Install the Arduino Bluetooth app from the Play Store.
2. Give the app microphone permission and enable Bluetooth.
3. Pair your device with HC-05.
4. Open the app and select Connect Robot from the menu.
5. Select the HC-05 module from the list of paired Bluetooth devices.
6. Program the Arduino.
7. Connect the Arduino to a 9v battery or computer.
8. Control your home using your phone.

- **The key components of this project are:**

- Arduino, Bluetooth module, Relay module, LCD display, Smartphone, Android app.

The app receives your voice command and sends it to the Bluetooth module wirelessly.

Arduino Code for Voice Controlled Home Automation

```
//voice Controlled Home Automation
#define relay1 2 //Connect relay1 to pin 2
#define relay2 3 //Connect relay2 to pin 3
#define relay3 4 //Connect relay3 to pin 4
#define relay4 5 //Connect relay4 to pin 5
String voice;
void setup()
{
  Serial.begin(9600); //Set rate for communicating with phone
  pinMode(relay1, OUTPUT); //Set relay1 as an output
  pinMode(relay2, OUTPUT); //Set relay2 as an output
  pinMode(relay3, OUTPUT); //Set relay3 as an output
  pinMode(relay4, OUTPUT); //Set relay4 as an output
  digitalWrite(relay1, HIGH); //Switch relay1 off
  digitalWrite(relay2, HIGH); //Switch relay2 off
  digitalWrite(relay3, HIGH); //Switch relay3 off
  digitalWrite(relay4, HIGH); //Switch relay4 off
}
void loop()
{
  while (Serial.available())
  {
    //Check if there is an available byte to read
    delay(10);
    //Delay added to make thing stable
    char c = Serial.read();
    //Conduct a serial read
```

```
    if(c == '#') {break;}|
    //Exit the loop when the # is detected after the word
    voice += c;
    //Shorthand for voice = voice + c
  }
  if (voice.length() > 0)
  {
    //Voice Command to ON Relay 01
    digitalWrite(relay1, LOW);
    //Relay 01 ON
  }
  else if(voice == "*tum on LED")
  {
    //Voice Command to ON Relay 02
    digitalWrite(relay2, LOW);
    //Relay 02 ON
  }
  else if(voice == "*tum on alarm")
  {
    //Voice Command to ON Relay 03
    digitalWrite(relay3, LOW);
    //Relay 03 ON
  }
  else if(voice == "*tum on fan")
  {
    //Voice Command to ON Relay 04
    digitalWrite(relay4, LOW);
    //Relay 04 ON
  }
  }
}
```

```

else if(voice == "*turn off light")
{
    //Voice Command to OFF Relay 01
    digitalWrite(relay1, HIGH);
    //Relay 01 OFF
}
else if(voice == "*turn off LED")
{
    //Voice Command to OFF Relay 02
    digitalWrite(relay2, HIGH);
    //Relay 02 OFF
}
else if(voice == "*turn off buzzer")
{
    //Voice Command to OFF Relay 02
    digitalWrite(relay3, HIGH);
    //Relay 03 OFF
}
else if(voice == "*turn off fan")
{
    //Voice Command to OFF Relay 02
    digitalWrite(relay4, HIGH);
    //Relay 04 OFF
}
}

else if(voice == "*turn all devices on")
{
    //Voice Command to ON all Relays
    switchallon();
    //All Relays ON
}
else if(voice == "*turn all devices off")
{
    //Voice Command to OFF all Relays
    switchalloff();
    //All Relays OFF
}
voice="";
//Reset the variable after initiating
}
}

////////////////////////////////////
void switchalloff()
//Function for turning OFF all relays
{
    digitalWrite(relay1, HIGH);
    digitalWrite(relay2, HIGH);
    digitalWrite(relay3, HIGH);
    digitalWrite(relay4, HIGH);
}
void switchallon()
//Function for turning ON all relays
{
    digitalWrite(relay1, LOW);
    digitalWrite(relay2, LOW);
    digitalWrite(relay3, LOW);
    digitalWrite(relay4, LOW);
}
}
}

```

Result Analysis

- Voice-controlled home automation uses voice commands to control household appliances. It can also activate home security and emergency systems.
- Arduino-based home automation uses Bluetooth to control electronic devices.
- Voice-controlled home automation can reduce:-
 - Human effort,
 - Save time,
 - Increase energy efficiency.
- Home automation allows us to control household electrical appliances like:-
 - Light,
 - Door,
 - Fan,
 - AC
 - Etc.
- It also provides home security and emergency system to be activated. Home automation not only refers to reduce human efforts but also energy efficiency and time saving.

Conclusion & Future Scope

- The Arduino is connected to the Bluetooth module, all the appliances can be controlled using the Arduino but it needs to be within a small distance for it to connect to the Bluetooth.

- **Disadvantage:**

- Since Bluetooth module is used, the range at which the home appliances can be controlled is reduced.

- A voice-controlled home automation system can use Arduino, relay modules, and Bluetooth technology to turn on and off home appliances.
- The system recognizes voice commands given by the user.

- **The system works by:**

- Receiving voice commands from the user
 - Sending the commands to Arduino via Bluetooth
 - Decoding the commands from the Bluetooth module
 - Sending commands to the Relays to control the home appliances
- The system can turn on and off various home appliances, such as lights and fans.

One disadvantage of the system is that the assistant may not be able to recognize commands properly.

- The future of home automation includes more advanced security features such as:

- Real-time monitoring

Main purpose of home automation system is to provide ease to people to control different home appliances with the help of the android application present in their mobile phones and to save

- Electricity,
- Time
- Money

➤ FUTURE SCOPE

- ✓ Internet of Things can also be combined with voice controlled home automation system.
- ✓ We can replace Bluetooth by GSM modem so that we can achieve device controlling by Sending SMS using GSM modem.
- ✓ We can also use Wi-Fi module instead of Bluetooth for home automation approach.
- Voice controlled home automation systems can be beneficial for people who are aged or physically disabled.
- These systems can allow users to control appliances in their homes using their voice.

The advantages of home automation include:

- Managing all home devices from one place
 - Flexibility for new devices and appliances
 - Maximizing home security
 - Remote control of home functions
 - Increased energy efficiency
 - Improved appliance functionality
 - Home management insights
- The future scope of voice controlled home automation systems includes:

- Building smart lights
- Sensing different phenomena such as temperature, water level, and humidity
- Using a transistor along with a relay to connect power to devices
- Sending commands to Arduino via Bluetooth HC-06

In addition, the future of home automation also includes more advanced security features such as facial recognition, biometric authentication, and real-time monitoring. This will provide homeowners with more secure and personalized control over their home.

References

Here are some references:

- Voice Controlled Smart Home Automation System
- Bluetooth And Arduino Uno-Based Voice-Controlled Home
- VOICE ASSISTANT HOME AUTOMATION USING ARDUINO

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